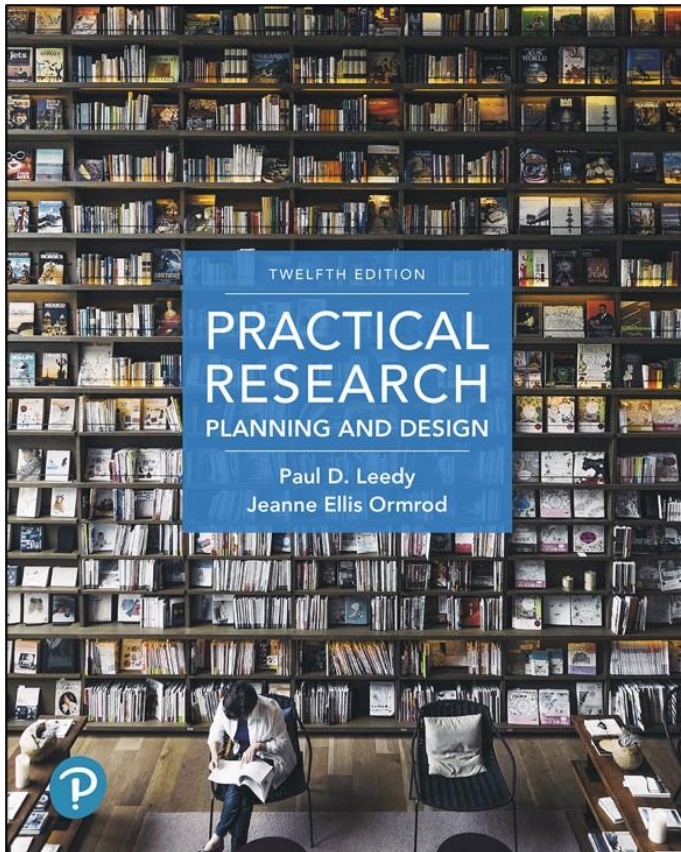


Practical Research: Planning and Design

Twelfth Edition



Chapter 1

The Nature and Tools of Research

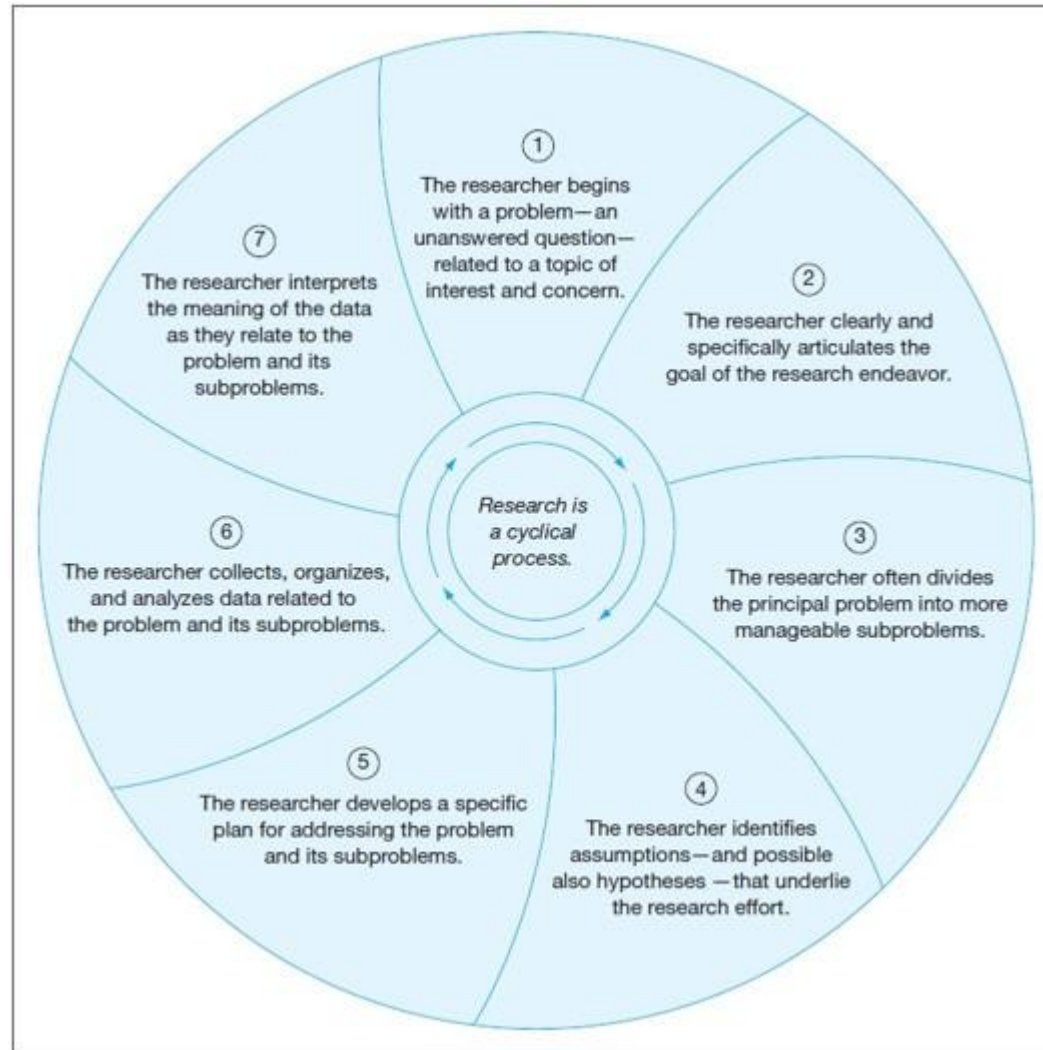
Research is (1 of 2)

- NOT merely:
 - gathering information.
 - rummaging around for hard-to-locate information.
 - transporting facts from one location to another.
- “Research” is a word that is used loosely in everyday conversation and therefore must be defined clearly for professional use.

Research is (2 of 2)

- A **systematic process** of
 - Collecting
 - Analyzing
 - Interpreting information (data)
- Goal: increase our understanding of a phenomenon about which we are interested or concerned
- Our focus is on **formal research**, or research intended to be communicated to a scientific community

Research is a Cyclical Process



Iterative Steps

- Identify a question or problem (unanswered question)
- Articulate a goal
- Divide the principal problem into more manageable subproblems
- Identify hypotheses and assumptions
- Develop a specific plan
- Collect, organize, and analyze data
- Interpret the data as they relate to the problem

Assumption

- An assumption is: Condition that is taken for granted.
- Two general assumptions that underlie most research projects:
 - The phenomenon under investigation is somewhat lawful and predictable.
 - Cause-and-effect relationships can account for certain patterns observed in the phenomenon.

Hypothesis (1 of 2)

- A hypothesis is:
 - A logical supposition
 - A reasonable guess
 - An educated conjecture
- Provides a tentative explanation for a phenomenon under investigation
- May direct thinking to possible sources of information necessary to resolve the research problem and its subproblems

Hypothesis (2 of 2)

- Hypotheses are not unique to research. Just imagine any time you've made a "guess" about the cause of some phenomenon.
- Hypotheses that are proposed prior to any data collection are called a **priori** hypotheses.

Types of Research

- Quantitative Research

- Involves looking at amounts, or quantities, of one or more variables of interest

- Qualitative Research

- Involves looking at characteristics, or qualities, that cannot be entirely reduced to numerical values

- Mixed Method Research

- Uses a combination of quantitative and qualitative methods

- Action Research

- Focused on outcomes for the practitioner

Philosophical Orientations Toward Research (1 of 3)

- Positivism
 - with appropriate measurement tools, scientists can objectively uncover absolute, undeniable truths about cause-and-effect relationships within the physical world and human experience.
- Postpositivism
 - true objectivity in seeking absolute truths can be an elusive goal and researchers bring biases to their work.

Philosophical Orientations Toward Research (2 of 3)

- Constructivism
 - no absolute truth — the “realities” researchers identify are simply human-created perceptions and interpretations.
- Phenomenology
 - focus is entirely on how human beings **experience** themselves and their world as they go through life.

Philosophical Orientations Toward Research (3 of 3)

- Pragmatism/realism
 - absolute truths may actually exist—even if they are exceedingly difficult to discover.
 - human beings' self-constructed beliefs about phenomena are legitimate objects of study in their own right.

Tools of Research

- Specific mechanisms or strategies used to collect, manipulate, or interpret data
 - The library and its resources
 - Computer technology
 - Measurement
 - Statistics
 - Language
 - The human mind
- Methodology dictates the tools the researcher selects

Library and its Resources

- Access to books, microforms, CDs, DVDs, online databases
- Fast and efficient means of locating and accessing information
- Access to library holdings around the world

Computer Technology is Used for

- Generating and organizing ideas
- Scheduling, structuring, and coordinating projects
- Finding literature
- Networking with others
- Storing and analyzing data
- Writing, editing, formatting
- Producing graphics
- Presenting results

Measurement

- Usually used in quantitative research
- More difficult for qualitative research
- Common instruments
 - Scales
 - Rulers
 - Stopwatches
- Specialized instruments
 - Telescope
 - MRI

Main Functions of Statistics

- Describe the data
 - Descriptive statistics summarize the general nature of the data obtained
- Draw inferences from the data
 - Inferential statistics help the researcher make decisions about the data

Language as a Tool

- Allows us to communicate and think more effectively
 - reduces complexity of the world
 - allows abstraction of the environment
 - enhances the power of thought
 - facilitates generalizations and drawing inferences

Communicating Effectively through Writing (1 of 2)

- Be specific and precise: Say exactly what you mean
- Keep your primary objective in mind at all times
- Provide an overview of what you will be discussing
- Organize ideas into categories by using headings and subheadings
- Use concrete examples to make abstract ideas more understandable

Communicating Effectively through Writing (2 of 2)

- Use figures and tables
- Regularly summarize what you've said
- Anticipate having to write multiple drafts
- Check your final draft
 - Grammar
 - Punctuation
 - Spelling

Using the Tools in Word-Processing Software (1 of 2)

- Beyond the basic tools of word-processing software, the following are also quite helpful:
 - Outlining
 - Setting headers and footers
 - Creating tables
 - Inserting graphics
 - Creating footnotes
 - Using international alphabets and characters
 - Tracking changes

Using the Tools in Word-Processing Software (2 of 2)

- General recommendations for using a word-processor effectively:
 - Save and backup your document frequently.
 - Use spell checker and grammar checker but don't rely on them exclusively
 - Print out a paper copy for final proofreading and editing

The Human Mind (1 of 5)

- The most important tool in the researcher's toolbox
- Used for critical thinking
- Evaluates
 - Accuracy
 - credibility
 - worth of information
 - lines of reasoning

The Human Mind (2 of 5)

- Evaluation may take a variety of forms:
 - verbal reasoning
 - argument analysis
 - probabilistic reasoning
 - decision making
 - hypothesis testing

The Human Mind (3 of 5)

- Deductive logic
 - Begins with one or more premises, statements or assumptions that the researcher initially takes to be true
 - Valuable for generating research hypotheses and testing theories
- Inductive reasoning
 - Begins with an observation (sample)
 - Observation then used to draw conclusions about entire classes of objects or events (population)

The Human Mind (4 of 5)

- Scientific method
 - Process of collecting & analyzing data systematically
 - Involves thinking actively and intentionally
 - Yields hypotheses
- Theory Building
 - An organized body of concepts and principles
 - Intended to explain a particular phenomenon

The Human Mind (5 of 5)

- Collaboration with other minds
 - Increases variety of
 - Perspectives
 - Backgrounds
 - Areas of expertise
 - May reduce bias

Exploring Research in Your Field

- Juried (or **refereed**) research report
 - Judged by experts in the field and deemed to be of sufficient quality and importance to warrant publication.
- Nonjuried (or **nonrefereed**) report
 - Appears in a journal or on the Internet without having been reviewed or selected by experts.
 - Some nonjuried reports are excellent, but others may not be.

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